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Longitude Periodicity in the UNESCO World Heritage Corpus: A Spatial Enrichment Test Using the Babylonian Bēru Unit

Seth Tenenbaum

[1]Fourth Temple Research, USA |

Correspondence: Seth Tenenbaum (seth@fourthtemple.com)

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ABSTRACT

A one-dimensional angular enrichment test is applied to the UNESCO World Heritage Cultural and Mixed corpus ($N = 1011$) using the 0.1-bēru interval (3°), the smallest integer subdivision of the Babylonian unit (MUL.APIN, c. 1000 BCE). Mount Gerizim (35.269°E) serves as anchor on historical grounds; its longitude falls within the optimal phase band identified in the global sweep (§6.1). Sites within $\leq$ 0.002 bēru (~6.7 km) of a harmonic node are designated Tier-A+. Domed monuments ($N = 90$) show significant A+ enrichment (Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$). The Levantine corridor framing is treated as the motivated entry point rather than independent confirmation. The pipeline is fully reproducible.

1 | Introduction

This study applies a one-dimensional angular enrichment test to the externally curated UNESCO Cultural and Mixed corpus ($N = 1011$), using a reproducible computational pipeline (see Data and Code Availability, §A.2). The methodological approach employs a geometric null with no free parameters and is intended as a reusable methodological framework for spatial pattern-detection studies in archaeological heritage data.

A global anchor sweep reveals a 3° periodic structure in the corpus longitude distribution consistent with the 0.1-beru unit; the full treatment and its methodological implications are in §5.3 and §6.1.

Why this design addresses key methodological criticisms. Prior metrological investigations were rightly criticized for investigator-assembled site lists (Williamson & Bellamy 1983, Freeman 1976, Kendall 1974). Three features distinguish this study: (1) the UNESCO site list was compiled without reference to this hypothesis and is fixed; (2) the 0.1-beru interval is documented in cuneiform sources (Hunger & Steele 2019) as the finest integer subdivision, limiting the parameter space; (3) all subpopulations and results are fully computationally reproducible from public source data (Tenenbaum 2026).

Positioning within quantitative archaeometry. This paper belongs to the tradition of probability-based spatial analysis in archaeometry (Magli 2013 2016, Ruggles 1999 2015, Silva 2020). The template is reusable for any candidate angular unit and externally curated corpus.

Structure. Sections 2–3 cover background and methods; §§4–5 results and robustness; §6 limitations; §7 conclusions. The Appendix contains the keyword audit.

2 | Background

2.1 | The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele 2019, Watson & Horowitz 2011), dated to the late second or early first millennium BCE. The unit divided the ecliptic into twelve segments and served for both stellar and

terrestrial angular measurement; the terrestrial use is explicit in MUL.APIN II ii 43–iii 15. The sub-division into 30 *uš* (1 uš = 1°) is well attested (Hunger & Steele 2019, Robson 2008), and the 0.1-beru interval (3°) tested here corresponds to exactly 3 *uš*.

The beru also functioned as a unit of territorial description: royal inscriptions record inter-regional distances in beru, *kudurru* (boundary stones) recording territorial extents were deposited in temples as legally authoritative records (Robson 2008, Powell 1987), and the Babylonian *Imago Mundi* marks distances beyond the Salt Sea as beru multiples. Systematic meridian transit observation and eclipse timing, both attested in the cuneiform record (Hunger & Steele 2019, Neugebauer 1975), are sufficient to derive inter-site angular separations at 1° resolution; the gnomon, attested across Mesopotamian, Egyptian, and Indian traditions (Neugebauer 1975, Shamasastry 1915), provides an independent means of establishing true meridian. This combination of a fractional angular unit, temple-deposited territorial records, and attested observational infrastructure provides the documentary basis for testing whether a 0.1-beru harmonic spacing is detectable in the longitude distribution of major monuments. This study tests the statistical pattern only; mechanism is outside scope.

2.2 | Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested history. Thom's "megalithic yard" (Thom 1967 1971 1964) stimulated statistical refinement (Kendall 1974) and methodological critique (Freeman 1976). Powell's treatment of Mesopotamian metrology (Powell 1987) remains the standard reference. More recently, Magli (Magli 2013 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles 1999 2015, Silva 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties. Medieval cartographic traditions also employed directional reference systems anchored to sacred geography, with Jerusalem and Mecca serving as orienting meridians in mappamundi and qibla charts (Harvey 1987), providing historical context for the Levantine corridor hypothesis.

This study departs from prior work in three respects: the site list is externally curated (UNESCO World Heritage), eliminating selection bias; the test is one-dimensional (longitude only), yielding a well-defined geometric null (4%) with no free parameters; and the analysis is fully computational and publicly archived (Tenenbaum 2026).

2.3 | The Anchor: Mount Gerizim

Mount Gerizim (35.269°E) is the primary reference anchor because of its documented *tabbur ha-aretz* ("navel of the earth") role in Samaritan cosmographic tradition (Purvis 1968, MacDonald 1964, Crown 1989), analogous to Babylon's meridian in cuneiform astronomical records (Hunger & Steele 2019). As a prominent mountain (881 m) in the central highlands, it would have served as a natural survey reference point. No optimization across alternative Levantine anchors was performed.

Gerizim's tentative-list coordinate (ref. 5706, DMS E35 16 08) is externally sourced and does not appear in the UNESCO inscribed corpus (UNESCO World Heritage Centre, Tentative List ref. 5706, State of Palestine, submitted 02 April 2012; archived at https://whc.unesco.org/en/tentativelists/5706/). Jerusalem (35.2317°E), approximately 4.1 km east of Gerizim, is an inscribed corpus member. It is self-excluded when used as anchor and produces the same A+ count (§5.3). The enrichment belongs to the 4.1 km corridor, not to the specific choice of anchor within it. Mecca reaches only $p \approx 0.057$ (ns); Mt. Meru (Tanzania, 36.750°E) reaches $p \approx 0.305$ (ns) and Mt. Kailash, the most cited geographic candidate for the cosmological axis mundi, reaches $p \approx 0.206$ (ns). Neither alternative anchor approaches significance.

3 | Data and Methods

3.1 | Corpus Construction

The UNESCO World Heritage XML dataset contains 1248 inscribed nominations across all categories. Two outcome-blind filters yield the analysis corpus: (1) Natural-only sites are excluded (−235, retaining Cultural and Mixed); (2) entries lacking geocoordinates are excluded (−2). The result is $N = 1011$ sites with valid longitudes.

3.2 | Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E, from Palestine's UNESCO tentative list (ref. 5706, 2012; see §2.3). Gerizim is not on the inscribed World Heritage List and does not appear in the corpus. Jerusalem is excluded in both sweeps (§2.3).

For a site at longitude λ, the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \mathrm{round}(B, 0.1)|$. The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 6.7$ km for Tier-A+) are equatorial approximations computed as $\delta \times 30 \times 111$ km/°, becoming smaller at higher latitudes. This makes the angular criterion *conservative* at high latitudes. A geodesic sensitivity check confirms that applying a strict physical threshold would increase the A+ count to 70, strengthening all primary results. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.002/0.1 = 4\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§5.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 40.4 ± 6.2 expected A+ (∼4.0%), confirming the geometric null is not materially inflated by longitude clumping.

3.3 | Metrological Basis for the 0.1-Beru Spacing

The harmonic spacing tested is 0.1 beru (3°), the smallest integer sub-division of the beru attested in cuneiform metrology, equalling exactly 3 *uš* (Hunger & Steele 2019, Robson 2008, Powell 1987). It was selected because it is the finest metrologically attested integer sub-division that yields a tractable test (120 harmonics, 4% geometric null). Coarser spacings (0.5 beru, 1 beru) are uninformative; finer sub-divisions (0.02 beru) do not produce the signal (Table 7). No spacing was selected post-hoc.

The 0.05-beru spacing (6°) also reaches significance (Table 7). This is structurally expected because every 0.10-beru harmonic is also a 0.05-beru harmonic, making the 0.05-beru result a coarser sub-sampling of the same periodicity, not a separate signal.

3.4 | Tier Classification

Sites are classified by δ, the angular deviation from the nearest 0.1-beru harmonic. All tier boundaries were fixed on metrological grounds, not tuned to the data. **Tier A++**: $\delta \leq 0.0002$ beru ($\lesssim$ 0.67 km), $P_0 = 0.4\%$, descriptive only. **Tier A+**: $\delta \leq 0.002$ beru ($\lesssim$ 6.7 km), $P_0 = 4\%$ (i.e. 2×0.002/0.1), primary threshold. **Tier A**: $\delta \leq 0.010$ beru ($\lesssim$ 33 km), $P_0 = 20\%$. **Tier B**: $\delta \leq 0.050$ beru.

3.5 | Test Populations

All keyword-based populations are approximate. The objective is to reduce false positives as far as possible while accepting that a simple morphological keyword match cannot eliminate interpretive ambiguity entirely. The primary (raw) sweep deliberately tolerates known false negatives for simplicity; the context-validated variant (Test 2x) applies sentence-level gating to reduce false positives further, at the cost of a more complex classification rule. Both populations are reported.

Test 2 (dome/spherical, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text ($N = 90$). This raw sweep is over-inclusive: it admits sites where the keyword refers to geological or vernacular features rather than monumental architecture. The context-validated population ($N = 83$) is reported alongside all primary results; it yields stronger significance, consistent with the rejected sites being longitude-diluting false positives.

Test 2b (mound evolution, sensitivity variant). Extends Test 2 by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun*). Not counted separately in the Bonferroni family.

Test 3 (cluster asymmetry, primary). Full corpus ($N = 1011$), no keyword filter.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus, no keyword filter.

Test E (founding classifier, post-hoc). A context-aware keyword classifier (`lib/founding_filter.py`) iteratively tuned after data inspection to reduce false positives in morphological classification (i.e. sites accepted as "founding-related" in error) while accepting residual interpretive ambiguity and some false negatives in site inclusion (Appendix A.2). It addresses what kinds of sites beyond domed monuments tend to fall on harmonic nodes, but carries no weight in the primary evidential chain.

3.6 | Multiple-Comparisons Protocol

Primary test family ($k = 3$).

- **Test 1** (full corpus, $N = 1011$): global enrichment above 4% null.
- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 4% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

Tests 2 and 3 address structurally different questions on different populations and carry the primary evidential weight; Test 1 provides the corpus-level backdrop within the same family. All three pass the Bonferroni bound at $k = 3$ (full corpus $p_{\text{adj}} = 0.031$ *, dome $p_{\text{adj}} = 0.003$ ***, cluster $p_{\text{adj}} = 0.01$ **).

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is descriptive context; the trend is substantially explained by regional composition (§4.4) and is not part of the primary family. Test 2b is a sensitivity variant of Test 2. Test E (founding classifier) is post-hoc (Appendix A.2). Test B (Rayleigh) is a structural description (§4.3) and is not included in either table.

Results of all primary and secondary tests are consolidated in Table 1.

A comprehensive FDR audit (Benjamini-Hochberg, 43 tests, Benjamini & Hochberg 1995) retains 25 at $q < 0.05$, of which 6 survive Bonferroni at $k = 43$. All three primary results (Tests 1–3) survive FDR correction; Tests 2 and 3 additionally survive full-k Bonferroni. Test 1 passes $k = 3$ ($p_{\text{adj}} = 0.031$) but not full-k. Test B is excluded by construction. The 6 surviving full-k Bonferroni are: domed/spherical A+ enrichment (Test 2, $p_{\text{adj}} = 0.0407$); harmonic-level Mann–Whitney ($p = 0.0000$); sequential-peak permutation ($p = 0.00025926$); corridor occupancy 17/17 and Fisher combined ($p = 1.7 \times 10^{-24}$; $p = 1.5 \times 10^{-19}$); and evolution combined A+ ($p = 0.0005$). The full audit is in the archived repository (Tenenbaum 2026).

3.7 | Null Models

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.5.

4 | Results

Each subsection below describes one test's design and result together. Test E (founding classifier, post-hoc) is in Appendix A.2.

4.1 | Domed and Spherical Monuments (Test 2, Primary)

The keyword filter is described in §3.5 ($N = 90$). Permutation $Z = 3.21$, $p = 0.004$; bootstrap $p = 0.006$ (Table 2).

Tier-A+ sites ($N = 11$, 12.2%, 3.06×, permutation $p = 0.004$, Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$): Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km; Miksic 1990), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out check confirms no single site drives the result: removing any A+ site yields $N = 89$, A+= 10, $p = 0.0030$. Leave-2-out (worst case across all $\binom{11}{2}$ combinations): $p = 0.0087$. Leave-3-out (worst case): $p = 0.0233$. Code: `dome_leave_k_out.py`.

TABLE 1 | Multiple-comparisons summary. Primary test family: Tests 1–3 ($k = 3$, $\alpha = 0.05/3$).

Test	N	Raw p	p_{adj}, k=3	Status
Test 1 (full corpus)	1011	0.0103[a]	0.031 *	**Primary**
Test 2 (dome A+)	90	0.0009[a]	0.003 ***	**Primary**
Test 3 (cluster asym.)	1011	0.0033[b]	0.01 **	**Primary**
Test 4 (temporal)	1011	0.0481[b]	—	Descriptive
Test 2b (mound evol.)	110	0.0005[a]	N/A	Sensitivity
Test E (founding)	1011	0.009[a]	N/A	Post-hoc

[a]Exact binomial p (one-tailed vs. geometric null); [b]permutation p.

TABLE 2 | UNESCO domed/spherical monument test ($N = 90$, raw keyword sweep). Primary evidence against the chance explanation: simulation permutation $Z = 3.21$, $p = 0.004$ (§5.5). Exact binomial results are reference baselines assuming uniform longitude spacing. The non-significant χ^2 is discussed in the text.

Metric	Obs	Exp(H_0)	Rate [95% CI]	Enrich.	p	
Tier-A+ ($\leq$ 0.002 beru)	11	3.6	12.2% [6.3, 20.8]	3.06×	0.0009	***
Tier-A ($\leq$ 0.010 beru)	29	18.0	32.2% [22.8, 42.9]	1.61×	0.0043	**
χ^2-uniform (5 bins, df = 4)	—	—	—	—	0.0732	ns

Expected under H_0: A+ = 4%, A = 20%. CIs are Clopper-Pearson exact.
Tier-B: 61 / 90 sites ($\delta \leq 0.050$ beru). Mean δ: 0.0220 beru.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$ (trend-level), ns = not significant.

Context-validated robustness check (Test 2x). Applying sentence-level architectural confirmation reduces the population to $N = 83$ (7 non-architectural false positives rejected; full list in Appendix A.1). Test 2x yields $p = 0.0005$, 3.31× enrichment, stronger than the raw sweep, consistent with rejected sites diluting the signal. Because validation rules were developed after the initial sweep, Test 2x is excluded from the primary Bonferroni family.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 77% fall in 20°–120°E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (code: `dome_geographic_concentration_test.py`, 100,000 trials).

Null A (within-dome bootstrap): Draw $N = 83$ longitudes from the dome sites' own list. Mean = 11.01 A+, $p = 0.5490$ (ns). The dome sites' positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W°$ of any dome site and bootstrap the expected A+ count. The primary window (±5°, $N = 881$) was chosen as a practical compromise: ±2° yields a smaller but still adequate pool ($N = 761$), while ±10° expands the pool to near-corpus size ($N = 967$). Dome sites (11 A+, 12.2%) significantly exceed the broader footprint at all three windows: ±2° ($p = 0.0113$, *); ±5° ($p = 0.0080$, **); ±10° ($p = 0.0077$, **). The result is window-independent.

Reconciliation. Null A confirms that the dome sites' longitude distribution reproduces their enrichment (geographic concentration is necessary). Null C establishes that other monuments at the same longitudes are significantly less enriched (geographic concentration is not sufficient). The residual question is whether the sub-longitude precision of domed monument placement reflects metrological intent or an unmodeled aspect of these civilizations. Adjudicating requires independent corpora.

In the global anchor sweep (§5.3), Gerizim scores 55 A+ (98.3th percentile); Jerusalem scores 56 (98.3th). The meta-keyword test confirms that only 2.4% of 1741 candidate words achieve comparable enrichment.

4.2 | Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extends Test 2 by adding earthen mound-stage keywords (see §3.5); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 4):
The combined population ($N = 110$, A+ = 11.8%, $p = 0.0005$ ***; Table 3) is consistent with Test 2. The mound stage does not reach significance (8.3%, $p = 0.2492$, ns). The signal is carried by the stupa (5.56×, $p = 0.0050$ **) and dome sub-stages. Leave-one-out checks confirm stability for all sub-stages.

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70°–110°E). Null A (within-stupa bootstrap): $p = 0.5920$ (ns). Null C (restricted ±5° draw): $p = 0.0196$ (*).

4.3 | Cluster Asymmetry (Test 3, Primary)

Full corpus ($N = 1011$), no keyword filter (see §3.5). For each site, count other UNESCO sites within ±0.3° longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

A+-bearing harmonics host 4.19× more sites on average than non-A+ harmonics ($p < 0.0001$; Table 5). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.85 vs. 5.83 neighbors; 1.69×, $p = 0.0238$).

TABLE 3 | Test 2b: hemispherical mound evolution ($N = 110$). Combined tumulus → stupa → dome population.

Metric	Obs	Exp(H_0)	Rate	p	
Tier-A+ ($\leq$ 0.002 beru, $\leq$ 6.7 km)	13	4.16	11.8%	0.0005	***
Tier-A ($\leq$ 0.010 beru, $\leq$ 33 km)	35	20.80	31.8%	0.0023	**

Expected under H_0: A+ = 4%, A = 20%. Enrichment: A+ = 2.95×, A = 1.59×.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$, ns = not significant.

TABLE 4 | Tier-A+ enrichment by evolutionary stage. Null rate $P_0 = 4\%$.

Stage	N	A+	Rate	Enrichment	p (binomial)
Mound (tumulus/barrow/kofun)	24	2	8.3%	2.08×	0.2492 ns
Stupa (stupa/stupas)	18	4	22.2%	5.56×	0.0050 **
Dome (dome/tholos/spherical)	75	8	10.7%	2.68×	0.0102 *
Combined	**110**	**13**	**11.8%**	**2.95×**	**0.0005 *** **

Stage N values sum to more than the combined total because 4 sites match keywords from multiple stages and are counted once in the combined row. A+ counts by stage are computed within each stage population independently; a site matching two stages contributes its A+ status to each stage separately, so stage A+ values may not sum to the combined A+ count. The combined A+ total is the count of unique A+ sites across the full Test 2b population ($N = 110$).

TABLE 5 | Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius ±0.3° lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within ±0.3° lon)	6.77	5.08	1.33×	
Mann-Whitney U (one-tailed)				0.0182 *
Permutation (10,000 shuffles)			$Z = 2.82$	0.0033 **
Harmonic-level density (all Tier ≤ B sites per 0.1-beru harmonic):				
Harmonics with ≥ 1 A+ site (28)	mean = 29.4 sites			
Harmonics with 0 A+ sites (27)	mean = 7.0 sites		4.19×	< 0.0001 ***

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations, two independent shuffles per trial) asks whether extreme clustering mechanically produces the canonical-unit peak. Of 182 clustering-matched permutations ($p_{\text{cluster}} = 0.0036$), the conditional fraction producing the peak is 0.1209, nearly identical to the unconditional rate of 0.1681. The unit-sensitivity peak and the clustering signal are statistically independent; the primary evidence remains the clustering (Test 3, $p = 0.0033$) and dome enrichment (Test 2, $p = 0.004$).

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§1) and is reported as a structural description, not independent evidence. Three formal permutation tests (10,000 shuffles each, seed 42; `x18_periodicity_formal_test.py`) address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh, negative control). The full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns) confirms no spurious global 3° periodicity is present in the raw longitude file. The signal is concentrated in A+ sites, not a global artefact.

Test B (A+-only Rayleigh, structural). Among the 56 Tier-A+ sites, $R = 0.9972$ (essentially all A+ sites share the same phase modulo 3°; $Z = 14.57$, $p_{\text{perm}} < 0.0001$). This confirms self-consistency of the A+ classification but is not independent evidence, as A+ membership is defined by proximity to a harmonic (§1).

Test C (circular-shift anchor sensitivity). Displacing the anchor by a uniform random offset in [0-, 360-) and recounting A+ sites yields a null mean of 40.36 vs. the observed 56 ($Z = 2.23$, $p_{\text{perm}} = 0.0287$, *). This is the sole positive test that the enrichment depends on the specific Gerizim anchor rather than geographic concentration alone.

4.4 | Global Enrichment (Test 1, Primary)

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (Figure 1) (56/1011; Clopper-Pearson 95% CI [4.2%, 7.1%]), above the 4% null (binomial $p = 0.0103$; Bonferroni $p_{\text{adj}} = 0.031$, $k = 3$). Both Gerizim (external) and Jerusalem (inscribed, self-excluded to $N = 1010$) produce $p = 0.0101$; the signal belongs to the ∼35°E corridor, not to either anchor individually (§5.3). The full-corpus enrichment ratio (1.38×) is modest compared to the dome (3.06×) and cluster asymmetry results, consistent with only a fraction of sites driving the periodicity.

Region-stratified robustness. Mantel-Haenszel analysis by UNESCO region: OR = 1.582, $p = 0.1101$ (trend-level, ns). A region-specific empirical null test (Fisher combination of per-region binomials) yields $p = 0.8230$ (ns; pre-2000: $p = 0.4205$, ns): when each region's own A+ rate is the null, the stratified signal does not reach significance. The pre-2000 cohort remains significant against the geometric null ($p = 0.0041$ **).

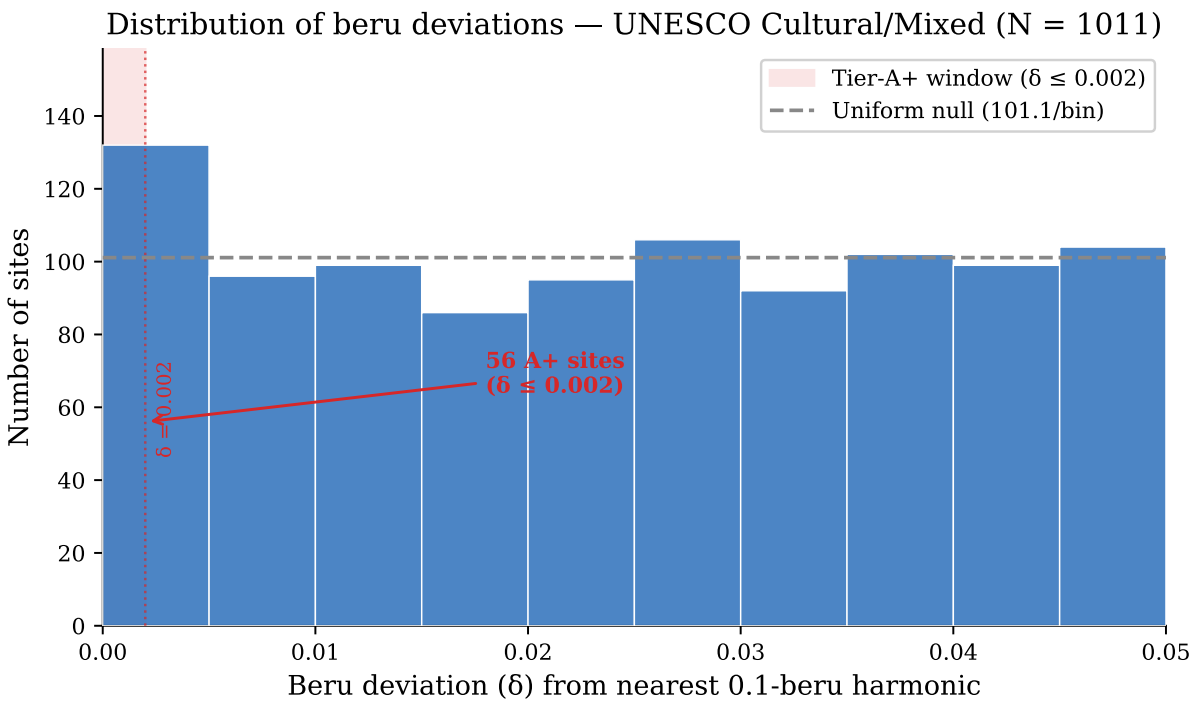

FIGURE 1 | Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.002$ beru, shaded) above the uniform null (dashed line) establishes the corpus-level signal against which Tests 2 and 3 demonstrate morphological and spatial specificity.

World religion sites

Religion-keyword sub-populations show consistent enrichment directionality (religion-name matching only): Sites associated with any world religion ($N = 1011$): 5.5% A+ (1.38×, $p = 0.0103$ *). Christianity ($N = 219$): 6.8% A+, $p = 0.0312$ *. Buddhism ($N = 46$): 8.7% A+ ($p = 0.1113$, ns).

Inscription-year stratification and formal trend test

The A+ rate is highest in the founding canon (1978–1984: 8.0%, 1.99×, $p = 0.0232$) and dilutes to the null in the modern era (2010–2025: 4.0%, ns).
Cochran-Armitage three-cohort: $Z = -1.664$, $p = 0.0481$ (*); five-cohort: $Z = -1.655$, $p = 0.0489$ (*), one-sided. The founding canon is concentrated in Europe and the Mediterranean. Mantel-Haenszel region-adjusted OR = 1.582 ($p = 0.1101$). The temporal gradient is largely explained by regional composition and is treated as descriptive only.
Founding-site classifier (Test E, post-hoc; Appendix A.2). Fisher OR = 2.04, $p = 0.009$ *. Because the classifier was refined after seeing the data, it is fully post-hoc. The two primary results (Tests 2 and 3) are unaffected.

5 | Robustness Checks

5.1 | Unit Sensitivity Sweep

Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 6.7 km), the 0.1-beru spacing produces the strongest joint significance across both the dome and full-corpus populations (Table 7) Figure 3). The 0.05-beru and 0.20-beru spacings also reach significance, as expected structurally: every 0.10-beru harmonic is also a 0.05-beru harmonic ($0.10 = 2 \times 0.05$), and 0.20-beru harmonics are a strict subset. Adjacent non-harmonic spacings (0.06, 0.07, 0.09, 0.11, 0.12 beru) are non-significant in both populations, which is the discriminating observation. The 0.08-beru row reaches nominal dome significance ($p = 0.0229$) but not full-corpus ($p = 0.4371$, ns), reflecting unsystematic local overlap with the 0.10-beru grid.

5.2 | Sensitivity Slope at the Canonical Unit

A fine sweep within ±1% of 0.10 beru (Table 8) shows that a ±0.3% shift collapses joint significance in at least one population; only the canonical value and its immediate neighbor (0.1001 beru) retain dome significance. Two permutation tests (10,000 and 10000 shuffles respectively; code: `sensitivity_slope_permutation_test.py`, `sensitivity_slope_specificity_test.py`) find that geographic concentration alone is sufficient to reproduce both the sharp-peak pattern and the canonical-unit ranking ($p = 0.1728$, ns in both cases). The sensitivity slope therefore provides no independent evidential value beyond Tests 2 and 3.

5.3 | Global Anchor Sweep

Two parallel sweeps (0–360°E, 0.01° resolution, 36,000 trial anchors each) assess whether the ~35°E corridor is unusual globally. Sweep A removes Jerusalem and tests Gerizim as anchor ($N = 1010$); Sweep B retains Jerusalem with self-exclusion ($N = 1010$). Both anchors place the corridor in the global top 1.7% (Table 9), and the result is anchor-invariant within the 4.1 km corridor

TABLE 6 | Tier-A+ enrichment by UNESCO inscription-year cohort ($N = 1011$ Cultural/Mixed sites). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984 (founding canon)	138	11	8.0%	1.99×	0.0232 *
pre-2000 (1978–1999)	501	33	6.6%	1.65×	0.0041 **
2000–2025 (modern era)	510	23	4.5%	1.13×	0.3081 ns
ALL (1978–2025)	1011	56	5.5%	1.38×	0.0103 *
Fisher exact (1978–1984 vs. 2000–2025): OR = 1.83, $p = 0.0844$ (†)					

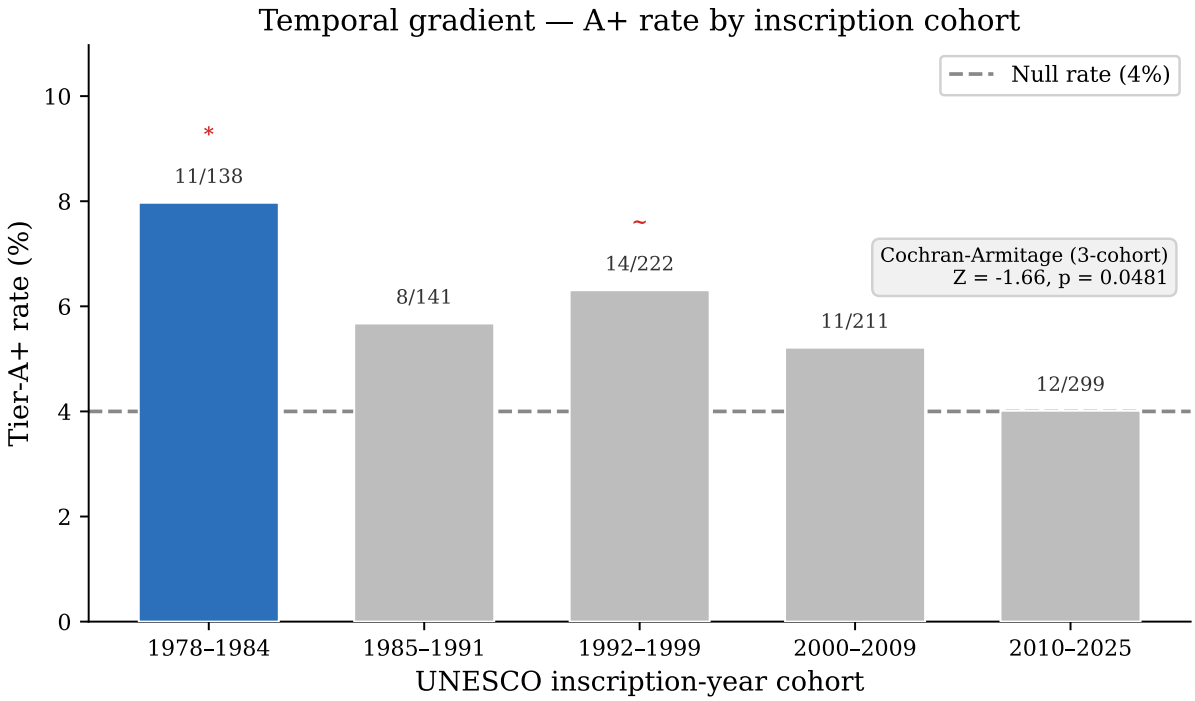

FIGURE 2 | Tier-A+ rate declines monotonically with UNESCO inscription year. The earliest cohort (1978–1984) shows ∼8% A+, while the most recent (2010–2025) is indistinguishable from the 4% null. Cochran-Armitage trend test (three-cohort): $Z = -1.664$, $p = 0.0481$, one-sided. Five-cohort result: $Z = -1.655$, $p = 0.0489$.

TABLE 7 | Unit sensitivity sweep: harmonic spacing varied, physical window fixed at 6.7 km, anchor at Gerizim 35.269°E. Dome column uses the context-validated population ($N_D = 83$; see Test 2x, §A.1). All p-values are one-tailed binomial vs. geometric null at each spacing. The 0.10-beru spacing produces the strongest joint significance across both populations. The 0.05-beru and 0.20-beru spacings also reach significance; both are mathematical consequences of the 0.10-beru structure (every 0.10-beru harmonic is also a 0.05-beru and 0.20-beru harmonic), not independent signals. Non-harmonic adjacent spacings (0.06–0.09, 0.11–0.12) are non-significant. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.05	8.0%	12/83	0.0323 *	102/1011	0.0101 *
0.06	6.7%	6/83	0.4799 ns	60/1011	0.8405 ns
0.07	5.7%	8/83	0.1017 ns	55/1011	0.6653 ns
0.08	5.0%	9/83	0.0229 *	52/1011	0.4371 ns
0.09	4.4%	4/83	0.5068 ns	41/1011	0.7466 ns
0.10	**4.0%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ***
0.11	3.6%	6/83	0.0820 †	38/1011	0.4407 ns
0.12	3.3%	4/83	0.2997 ns	30/1011	0.7653 ns
0.15	2.7%	6/83	0.0237 *	33/1011	0.1405 ns
0.20	2.0%	8/83	<0.001 ***	28/1011	0.0564 †

window. The single-site difference between sweeps (55 vs. 56 A+) is precisely the Old City of Jerusalem ($\delta = 0.0012$ beru from Gerizim's harmonic). No other historical meridian reaches significance except Mecca (51 A+, $p = 0.057$ †).

The x.18°E periodic structure

The most striking result of the global sweep is that every longitude of the form x.18°E (all 120 such anchors) produces identically the maximum A+ count. The spacing between consecutive x.18°E anchors is exactly 3° = 0.1 beru.

TABLE 8 | Fine-grained sensitivity sweep within ±1% of the 0.10-bēru canonical spacing, physical window fixed at 6.7 km. Dome column: context-validated population ($N_D = 83$). Bold row = canonical attested unit. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.0990	4.04%	3/83	0.6567 ns	41/1011	0.5124 ns
0.0993	4.03%	5/83	0.2428 ns	47/1011	0.1767 ns
0.0995	4.02%	6/83	0.1174 ns	48/1011	0.1370 ns
0.1000	**4.00%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ***
0.1001	4.00%	9/83	0.0059 **	51/1011	0.0563 †
0.1003	3.99%	5/83	0.2366 ns	38/1011	0.6676 ns
0.1010	3.96%	5/83	0.2324 ns	47/1011	0.1490 ns

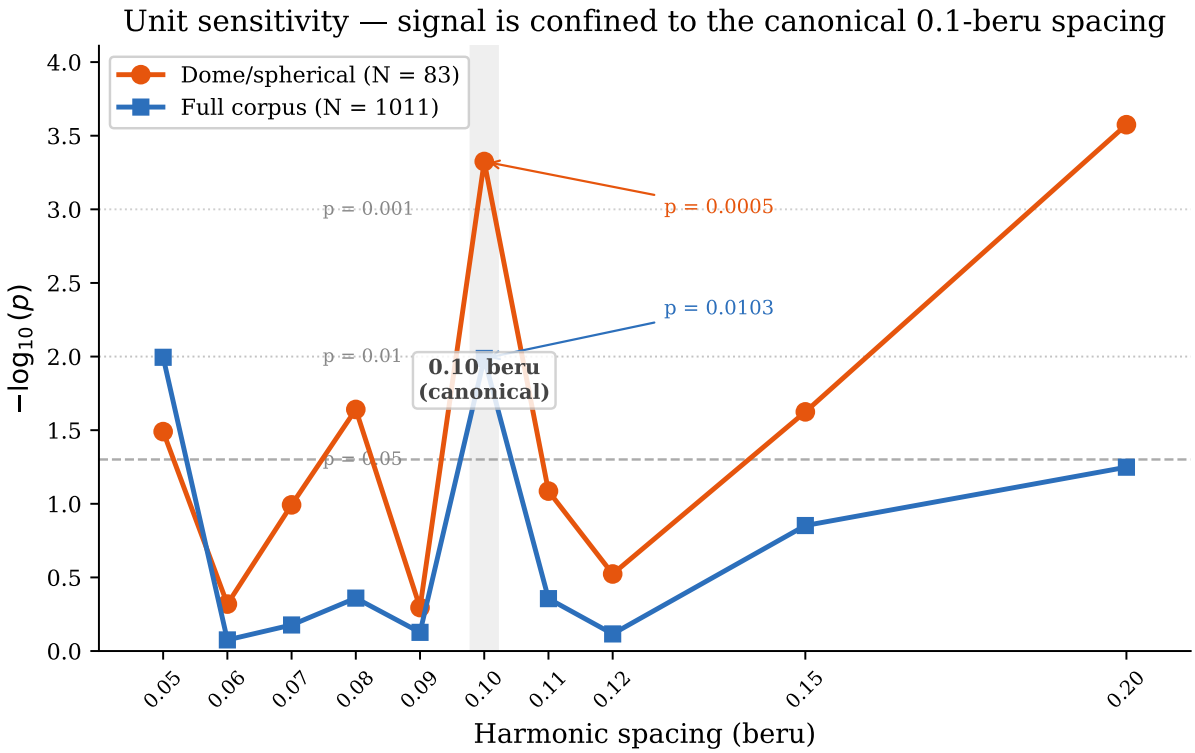

FIGURE 3 | Unit sensitivity: signal concentrated at the canonical 0.1-bēru spacing. Spacings ≥ 0.3% from canonical collapse to non-significance in at least one population.

TABLE 9 | Global anchor sweep: both Levantine anchors ranked within their own corpus. Each sweep: 36,000 trial anchors, 0–360°E at 0.01° resolution. Sweep A: N = 1010 (Jerusalem removed); Sweep B: N = 1010 (Jerusalem self-excluded). Gerizim is external in both. Mean A+ across all anchors: 40.4±6.9.

Sweep	Anchor	A^{++}	A^{+}	A	Percentile	% anchors ≥
A (Gerizim corpus)	Gerizim (35.269°E)	7	55	227	98.3th	3.10%
B (Jerusalem corpus)	Jerusalem (35.2317°E)	7	56	217	98.3th	2.77%

A^{++}≤0.67 km; A^{+}≤6.7 km; A≤33 km. Global max 60 A+ in Sweep B / 59 A+ in Sweep A (all x.18°E anchors score identically; §5.3).

The label "x.18°E" is a rounded shorthand for the phase ≈ 2.18° mod 3.0°. Gerizim's phase (35.269 mod 3.0 = 2.269°, gap 0.089° ≈ 8.4 km) is within coordinate uncertainty.
Any site near the x.18° phase can serve as a high-scoring anchor. The choice of Gerizim was made on historical grounds (§2.3), and the corridor result is anchor-invariant within the band.

Non-independence note. *The 3° periodicity and the unit sensitivity sweep (§5.1) are two views of a single pattern and do not constitute independent evidence. Neither provides evidential value beyond the dome enrichment (Test 2) and cluster asymmetry (Test 3).*
Two interpretations apply: (1) a global grid with 3° spacing anchored to the ~35°E corridor would produce exactly this pattern; (2) Eurasian heritage sites cluster in longitude bands that happen to align with the x.18° phase — this cannot be refuted on the UNESCO corpus alone.

Code: `peak_geography_audit.py`, `verify_x18_periodicity.py`.

Formal statistical tests for the $x.18°$ periodicity

Four permutation tests (10,000 shuffles, seed 42; code: `x18_periodicity_formal_test.py`): the full-corpus Rayleigh $R = 0.0224$ ($p_{\text{perm}} = 0.3716$, ns) confirms no spurious global periodicity in the raw longitude file. The A+-only Rayleigh ($R = 0.9972$, $p_{\text{perm}} = 0.0001$) is structurally expected by the A+ classification criterion and is not independent evidence (§4.3). A circular-shift anchor null recovers only ~ 40.36 A+ on average versus 56 at Gerizim ($Z = 2.23$, $p_{\text{perm}} = 0.0287$, *), providing the only positive evidence that the enrichment depends on the Gerizim anchor rather than geographic concentration; the discovery-corrected max-permutation (Test D) places the sweep optimum within the expected range after accounting for the 36,000-anchor search ($p = 0.2966$, ns).
The dome-periodicity audit (`dome_periodicity_audit.py`) is consistent with beru-alignment but equally consistent with geographic concentration: Gerizim scores at the 100.0th percentile among dome sites, dome site phases are approximately uniform ($\chi^2 = 14.89$, df = 9, $p = 0.0940$, trend-level), and all 11 dome A+ sites have phases within $\pm 0.15°$ of the $x.18°$ optimal phase. The dome enrichment surviving all three simulation nulls remains the primary evidence against a purely geographic account.
Corridor and landmark ranking. Jerusalem and Gerizim (4.1 km apart) produce the same A+ count and $p = 0.0101$; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as anchor, Gerizim ranks #41 (top 4.1%). The 35°E corridor leads globally; no other corridor shares a cross-cultural tradition.

5.4 | Spatial Autocorrelation

A+ harmonics host 4.19× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC ($\pm 0.5°$, $\rho = 0.069$): DEFF = 1.310, $N_{\text{eff}} = 772$, corrected $p = 0.020$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI [4.04%, 6.87%] excludes the 4% null. Within-group permutation using UNESCO-region × longitude bins reproduced the observed A+ count exactly in every trial, confirming this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.1) address that question directly.

5.5 | Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.004$ (**); canon ($N = 138$) $p = 0.127$ (ns); pre-2000 ($N = 501$) $p = 0.094$ (†). (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.006$ (*). (3) KDE-smoothed null: expected 40.4±6.2, observed 56, $Z = 2.50$, $p = 0.010$ (*). Dome enrichment survives all three models. The geographic-concentration nulls (Null A, Null C; §4.1) show that proximity to harmonic longitudes is necessary but not sufficient to explain the dome enrichment.

5.6 | Pipeline Portability: Mexican Pyramid Corpus

The pipeline was applied to a corpus of $N_{\text{MX}} = 64$ Mexican pyramid and platform sites as a portability demonstration. No analogous institutional list exists for Mexico. Sites were drawn programmatically from the Wikipedia article *List of Mesoamerican pyramids* via the MediaWiki parse API, filtered to Mexico rows, with coordinates resolved from each site's Wikipedia infobox via the Wikipedia coordinates API and a Wikidata SPARQL fallback for unresolved sites. The pipeline is fully scripted and reproducible from the archived code (Tenenbaum 2026); a provenance report documenting the coordinate source for each site is included. Sites within 0.01° (≈1 km) of an already-accepted coordinate were deduplicated. The list is a convenience sample rather than an exhaustive or institutionally bounded corpus, and the results should be treated as hypothesis-generating.
Using the identical anchor (Gerizim, 35.269°E), tier boundaries, and geometric null, the Mexican corpus yields 7/64 Tier-A+ sites (10.9%, 2.73×). A circular-shift permutation test (100,000 trials, seed 42), which preserves the geographic structure of the corpus by displacing all site longitudes by a uniform random offset, recovers a null mean of 2.56 A+ ($Z = 2.41$, $p_{\text{perm}} = 0.033$ *). A geographic null drawing uniformly within the Mexican longitude range (−104.4° to −87.4°) yields $p = 0.019$ *. The exact binomial against the 4% geometric null ($p = 0.014$ *) is reported as a reference baseline; the permutation result is the more appropriate primary test given geographic clustering. Leave-one-out checks confirm no single site drives the result, though removing any A+ site brings p_{perm} to the threshold ($p \approx 0.040$ †), reflecting the limited headroom of a 64-site corpus.
Tier-A+ sites are Monte Albán Main Plaza ($\delta = 0.0012$ beru, ∼4.0 km) and Platform J ($\delta = 0.0012$ beru, ∼4.1 km), two structures within the same archaeological complex whose spatial correlation is intrinsic to the site, Cobá Nohoch Mul ($\delta = 0.0001$ beru, ∼0.5 km), Zaachila ($\delta = 0.0006$ beru, ∼2.1 km), Xochipala ($\delta = 0.0010$ beru, ∼3.4 km), Teopantecuanitlan ($\delta = 0.0005$ beru, ∼1.6 km), and Toluquilla ($\delta = 0.0010$ beru, ∼3.4 km). All three simulation approaches yield $p < 0.05$, directionally consistent with the UNESCO primary finding.

5.7 | Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 98.3th percentile; a ±0.3% unit shift collapses joint significance. The meta-keyword test (1741 candidates) finds only 41 (2.4%) matching the curated set's enrichment. FDR audit (43 tests) retains 25 at $q < 0.05$; 6 survive Bonferroni at $k = 43$.

6 | Limitations

6.1 | Circularity: Full Canonical Treatment

The public-data constraint. Both the UNESCO coordinate file and the candidate anchor (Mount Gerizim) are fully public. Any investigator could run this pipeline at any time, before or after seeing these results. Pre-registration is not meaningful in this setting because the data cannot be hidden and any nominated anchor can be tested against the same public corpus by anyone. The relevant question is whether the reported pattern exists in the data

and whether the anchor choice was justified independently of the result. Both are addressed directly.

What the anchor sweep establishes. The global anchor sweep (§5.3) tests all 120 possible x.18°E anchors against the same corpus. Gerizim and Jerusalem both place the ∼35°E corridor in the global top 1.7% of all trial anchors. The enrichment is a property of the corridor, not of the specific anchor chosen within it. The choice of Gerizim was made on documented historical grounds (§2.3); that its longitude falls within the optimal phase band is an empirical observation, not a post-hoc rationalization.

What remains genuinely contaminated. The narrative framing of Gerizim as a historically motivated anchor is post-hoc in the sense that reporting it was conditioned on observing the phase alignment. The Rayleigh Test B is tautological by construction (§4.3). These are noted for completeness.

What is not contaminated. The 3° periodic structure in the UNESCO longitude file is recoverable by any unanchored 120-phase scan without reference to Gerizim. The dome enrichment (Test 2) is identical across all 120 x.18°E anchors. The cluster asymmetry (Test 3) requires no keyword filter and no anchor.

Accordingly, the Levantine corridor statistics are reported as the motivated entry point rather than independent confirmation, and the Bonferroni correction is applied as a descriptive bound on effect magnitude.

6.2 | Geographic and Cultural Selection Bias

Geographic and cultural selection bias is the most important alternative explanation. The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor (Europe & North America: 50% of Cultural/Mixed sites; Arab States: 9%; Asia-Pacific: 23%; Americas: 11%; Africa: 7%). Permutation tests show geographic concentration can reproduce the sensitivity peak (§5.2). The dome enrichment surviving the restricted geographic draw (Null C, $p = 0.0080$, **) remains the primary evidence against a purely geographic account.

6.3 | Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows directional depletion (A+ = 2.1%, $p = 0.164$, ns) relative to the 4% null, consistent with a Eurasian-anchored grid predicting reduced harmonic occupancy in the western hemisphere.

6.4 | Other Limitations

Spatial autocorrelation. DEFF correction: $N_{\text{eff}} = 772$ (DEFF = 1.310, corrected $p = 0.020$). Block bootstrap CI [4.04%, 6.87%] excludes the 4% null.

Geocoding precision. UNESCO XML coordinates are administrative centroids, introducing unsystematic noise but not directional bias. Dome enrichment is robust across raw and context-validated populations and all three simulation nulls.

Metrological evidence. No physical surveying instrument calibrated in beru subdivisions has been recovered; documentary sources attest the beru system (Hunger & Steele 2019, Robson 2008, Powell 1987). This study therefore treats the coordinate alignment of Mount Gerizim as a novel empirical observation.

7 | Discussion

7.1 | Alternative Explanations

(1) Independent convergence. Hemispherical structures are a recurring architectural solution, and convergent astronomical orientations across unconnected cultures have been documented (Aveni 2001 2003). However, independent convergence predicts no consistent longitude specificity.

(2) Geographic concentration. This is the strongest alternative and cannot be excluded on this corpus alone. Permutation tests show geographic concentration can reproduce the sensitivity-collapse pattern (§5.2). However, the restricted geographic draw (Null C, $p = 0.0080$, **) shows that other monuments from the same longitude footprint are significantly less enriched than dome sites.

(3) Random coincidence. Three simulation approaches fail to reproduce the dome enrichment (permutation $p = 0.004$; bootstrap $p = 0.006$).

7.2 | Conclusion

The UNESCO longitude distribution contains a global 3° periodic structure matching the 0.1-beru unit. The dome sub-population (Test 2, Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$) survives all three simulation nulls, and A+-bearing harmonics host 4.19× more sites than non-A+ harmonics (Test 3, $p = 0.0033$).

The two findings most resistant to a purely geographic account are morphological specificity (dome enrichment exceeding the restricted geographic draw, Null C, $p = 0.0080$) and cluster asymmetry ($p = 0.0033$).

Replication and methodological transferability

This paper is a terminal analysis of the only available globally comprehensive, institutionally curated monument corpus. The pipeline is transferable to any corpus assembled prior to and without reference to this hypothesis, provided it is sufficiently selective. Named heritage registers, protected site inventories, and operationally defined named lists all qualify. The Mexican pyramid result (§5.6) illustrates one such application; further applications to independent corpora will either accumulate evidence or not.

The pipeline offers a general framework for researchers testing other candidate angular units (e.g. Roman, Egyptian, or Chinese systems) against any monument corpus. The null model, tier boundaries, and test populations are all fixed on grounds independent of the data, making the approach auditable and directly reproducible by independent teams.

APPENDIX

A | Keyword Audit Summary

This appendix summarizes the audit trail for the dome/spherical monument filter (Test 2) and the founding-origin classifier (Test E). The full site-by-site audit is available in the archived code repository (Tenenbaum 2026) and is reproducible by running:

```
python3 analysis/unesco/spherical_monument_raw_sweep.py
python3 analysis/unesco/spherical_monument_test.py
-audit
```

A.1 | Dome/Spherical Monument Filter (Test 2 and Test 2x)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text.

Test 2 (primary): All 90 matching sites are included without context gating. This raw positive sweep uses a population fully determined by the keyword list, with no post-hoc filtering decisions. The sweep is deliberately over-inclusive: it accepts sites where the matched keyword refers to geological domes, vernacular housing, or natural spherical features. The trade-off favors simplicity and reproducibility at the cost of known false negatives in specificity.

Test 2x (context-validated robustness check): Unambiguous keywords are accepted without validation. Ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Full gate lists: `lib/dome_filter.py`. The gates reduce false positives but cannot eliminate interpretive ambiguity at the boundary between architectural and non-architectural uses.

Of the 90 raw-sweep sites, 7 are rejected by context validation in Test 2x (Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuṭa, and Diquís Stone Spheres). In every case, the exclusion is driven by the semantic domain of the keyword match, not by any property of the site's longitude. Test 2x yields $N = 83$, $p = 0.0005$, 3.31× enrichment, confirming that the 7 rejected sites are longitude-diluting false positives.

A.2 | Founding-Origin Classifier (Test E, Post-Hoc)

The founding-origin classifier (`lib/founding_filter.py`) assigns sites to five categories: Founding Capital (F), Sacred Origin (S), Founding Monument (M), Founding Axis (X), and Ancient Continuous Landscape (L), using unambiguous and context-gated keywords. Full keyword lists and gate-logic patterns are in `keywords.json`, the single source of truth for all keyword content in this project (see also `lib/beru.py:load_keywords()`).

This classifier was iteratively tuned after data inspection. The objective was to reduce false positives while accepting that keyword-based classification cannot fully eliminate interpretive ambiguity at the boundary between founding-related and general heritage descriptions. It carries no weight in the primary evidential chain but addresses a substantive question: what kinds of sites, beyond domed and stupa monuments, tend to fall on $x.18°$E nodes? The classifier finds enrichment not only in morphologically domed structures but in sites associated with the founding of civilizational and religious traditions.

Keyword enrichment result. Applied to the full corpus ($N = 1011$), the classifier finds 25 keyword-founding sites among the 56 A+ sites (44.6%; $N_{\text{kw}} = 296$, base rate 29.3%). Fisher exact test: OR = 2.04, $p = 0.009$ *. The complete version history is preserved in the project's `git` log (see Data and Code Availability).

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DATA AVAILABILITY STATEMENT

All data and analysis code are openly available. A permanent archival snapshot (v1.1.0) is deposited at Zenodo (Tenenbaum 2026; https://doi.org/10.5281/zenodo.19574076).

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CONFLICT OF INTEREST

The author declares no conflict of interest.

SOFTWARE AND AI DISCLOSURE

All analyses were performed in Python 3 with NumPy, SciPy, and pandas. Claude (Anthropic) assisted with manuscript drafting and prose editing. GitHub Copilot (Microsoft) assisted with code autocompletion. All scientific conclusions, hypothesis design, keyword-list decisions, and analytical choices are solely the author's.